

Day 2:

List `[]`

tuple `()`

dictionary

List:

it's order and mutable

Allow duplicates

`[]` → brackets use

tuple → immutable ~~{ }~~ → parentheses `()`

$m_1 = 35$

$m_2 = 56$

$m_3 = 75$

marks = `[35, 56, 75]`

output :

`[35, 56, 75]`

a = [1, 2, 3, 4, 5] → list
type(a)
output:
list

b = (1, 2, 3, 4, 5) type(b)
output:
tuple

INDEXING:

0 start 3 4
a = [1, 2, 3, 4, 5]

a[1]

output:
2

a[0] = 87
output:

a [2, 4, 87, 4, 5]

-5 -4 -3 -2 -1
[1, 2, 3, 4, 5]

output:
5

a[-4]
output:
2

a[-2] = 44

output:
[1, 2, 3, 44, 5]

Function:

- * Set of task perform Function
- * reusable block of specific code
- * repeated code avoid.

a.append → add at the end

a.append(9)

a [2, 4, 87, 44, 1, 9]

a.append(23)

a [2, 4, 87, 44, 1, 9, 23]

a.pop → last number remove

a.pop()

a [2, 4, 87, 44, 1, 9]

a.pop(4)

output

[2, 4, 87, 44, 9]

1 will be remove

planets = ["mercury", "venus", "Earth", "mars",
"Jupiter", "saturn", "Uranus",
"neptune"]

planets[2]

output:
earth

Slicing:

* Portion of slice.
* slice the portion and access.
start, stop, step
0th ↓ L > number of position
index exclusive move each time.
include
inclusive

Expression:

a[:]: Copy the entire sequence

a[1:4] elements from index 1 to 3

a[:3] First 3 elements

a[3:] From index 3 to end

a[::2] every second element

a[::-1] reverse the sequence.

planet [0:3] → elements 0 to 2
before

output:

['mercury', 'venus', 'earth']

planets[:3]

['mercury', 'venus', 'earth']

planets[2:5] start 2 to end 4

['earth', 'mars', 'Jupiter']

planets[1:-1]

-1 excluded so without neptune

all planets come.

planets[1:-4]

output:

['venus', 'earth', 'mars']

planets[-4:]

['jupiter', 'saturn', 'uranus', 'neptune']

planets[::2] → every 2nd element

output:

['mercury', 'earth', 'jupiter', 'uranus']

planets[:: -3]

output: → every 3rd element last start

['neptune', 'jupiter', 'venus']

planets[:: -1] → Reverse the Sequence

output:

Full planet Reverse.

a = [1, 2, 3, 5, 55, 66, 43]

len[a]

output:

7

Sum[a]

output:

add all num

Max[a]

output

66

min[a]

output:

1

Syntax:

#def function name()

write

return

def → Function define.

function name → not stored in variable

() → parameters

write → write the condition

return → Result. [10, 20]

def kaadhi(a, b):

c = a + b

return c

output:

30

[Resume block of code]


```
def Aarthi(a,b):
    c = (a/b) + (a*b)
    return c
```

Return & print → different

```
def aarthi(a):
    if a > 50:
        print("bias")
    else:
        print("no bias")
```

aarthi(22)

output:

no bias.

average([2, 4, 6, 3, 7])

Syntax:

```
def average(a):
    avg = sum(a) / len(a)
    return avg
```

output
4.4

Div:

```
def div(a,b,c):
    out = (a/b) + (b/c)
    return out
```

div(2,6,4)

Output:

1.833333

qn:

- * 3 section - A, B, C
- * Highest and average Find
- * winner and Highest